

Dancing with your Thumbs User Manual

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Overview

Dancing with your Thumbs is a simple, yet fun, spin on the classic rhythm game *Dance Dance Revolution*. The core gameplay loop is simple. The 16x16 LED board is broken up into four 16x4 columns. The top three lines are colored green, yellow, and green in order from the top. In each column, a line of red lights will appear at the bottom of the column at random intervals. This line will then move up the column until it goes off the top. While it overlaps with the top three lines, it overrides them, such that that line is red until the red line moves upward. For example, if the red line travels into the top row (normally green), that row becomes red until the line moves on.

Each column is attached to a button, corresponding to its position on the LED board. From left to right, the columns are matched to KEY[3], KEY[2], KEY[1], and KEY[0]. If a column currently has a red line on it and its corresponding button is pressed, the light will vanish, and the player will get points based on the location of the line. If the line is second from the top (these are the lights that are normally yellow) the player gets 2 points. If it's on the top or the third from the top (normally green), the player gets one point. If the line is anywhere else, or if the line goes off the top, the player will lose two points. The player's total score is shown on the HEX 7-segment displays, from 0 to 9999. It will not go below 0, nor above 9999 – there is no overflow.

The player will know if they timed a button press right by observing the top three rows. If they got a 'good' timing, all 3 will become green in that column. If they got a 'perfect' timing, all 3 will become yellow. If they missed, the rows won't change. No new lines of red lights will appear in a given column if there is already a line in that column. Similarly, no new lines will appear for the duration of the 'flash' upon getting a good or a perfect.

Dancing with your Thumbs has 4 different difficulty settings – easy, medium, hard, and extreme. The difficulty is based on the current state of switches 7 and 8. If both are 0, it's easy. If both are 1, it's extreme. If only 7 is on, it's medium, and if only 8 is on it's hard. Difficulty affects the game speed, with higher difficulty causing lights to spawn in and move faster. Finally, switch 9 is used to reset the game, and should be enabled upon every difficulty change.

Operation

Dancing with your Thumbs is simple to operate. It's built for the DE1-SoC and requires the optional 16x16 LED grid attachment. After plugging the grid in and turning the board on, load the design. If the top three rows light up green-yellow-green, the design has loaded properly. Turn switch 9 (reset) on for a moment, then turn it back off. The game should begin.

To change the difficulty, simply adjust switches 7 and 8 to your desired speed. Changing difficulty mid-game shouldn't have any adverse effects on the system, but it is recommended that you reset the game after doing so just in case.

Block Diagram(s)

The block diagrams continue downward, with each further diagram getting progressively more detailed. Modules with many complex connections have their connections abbreviated, these connections are expanded upon in the subdiagrams for their submodules. With the exceptions of `clk_counter` and `clock_divider`, clocks/enable signals are implicit. Reset is also implicit. Diagrams are presented roughly in order.















